# Kunal Pillai

## Mechanical Design Engineer

RESUME_1915

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Total Experience: 11 years

## PROFESSIONAL SUMMARY

Mechanical Design Engineer with 11 years of experience in cad design, using SolidWorks, CATIA, AutoCAD, GD&T. Experienced in design validation, tolerance analysis and measurable delivery outcomes.

## TECHNICAL SKILLS

Skills: SolidWorks, Prototype Testing, FEA, Sheet Metal, CATIA, 3D Modeling, GD&T, DFM, PLM, ANSYS, AutoCAD, Tolerance Stack-up

Methods: Design validation, Tolerance analysis, Manufacturability

## PROFESSIONAL EXPERIENCE

**Lead Mechanical Design Engineer | NovaPeak Engineering | Jan 2022 - Present**

- Released 113 controlled part drawings with complete GD&T definitions
- Reduced component mass by 14% while preserving required factor of safety
- Cut prototype iteration time by 20% using parametric CAD and standardized drawings
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

**Mechanical Design Engineer | Ironleaf Engineering | Jan 2019 - Dec 2021**

- Cut prototype iteration time by 41% using parametric CAD and standardized drawings
- Reduced assembly fit-up corrections by 19% through tolerance stack-up analysis

**Associate Mechanical Design Engineer | OrionPath Technologies | Jul 2015 - Dec 2018**

- Released 22 controlled part drawings with complete GD&T definitions
- Reduced component mass by 31% while preserving required factor of safety

## PROJECTS

**Pump Housing Validation | Tolerance Stack-up, PLM, Sheet Metal, Prototype Testing**

- Compared FEA stress predictions with prototype load-test results.
- Documented acceptance criteria and evaluated the design using tolerance analysis and manufacturability.

**Modular Conveyor Assembly | AutoCAD, Sheet Metal, 3D Modeling, FEA**

- Designed frame, transmission and tolerance-controlled mounting parts.
- Documented acceptance criteria and evaluated the design using tolerance analysis and manufacturability.

## EDUCATION

B.Tech in Mechanical Engineering, Silverlake School of Engineering, 2015
Relevant Methods: Design validation, Tolerance analysis, Manufacturability

## CERTIFICATIONS

SolidWorks Professional Training (2024)